Jialin Song

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Education

Simon Fraser University , Burnaby, BC, Canada — Aug 2024 - May 2028 (Expected)
Ph.D. in Computing Science

University of California Berkeley , Berkeley, CA, U.S. — Aug 2021 - May 2022
M.Eng. in Electrical Engineering and Computer Sciences

Worcester Polytechnic Institute , Worcester, MA, U.S. | *with high distinction* — Aug 2017 - Dec 2020
B.S. in Computer Science and Robotics Engineering (Double Major)

Experience

Microsoft Research , Redmond, WA — May - Aug 2026, May - Sep 2025 (Full-time), Oct 2025 - Apr 2026 (Part-time)
Research intern in the Deep Learning Group and Catalyst Lab, working on LLM jailbreak and agentic safety

International Computer Science Institute (ICSI) , Berkeley, CA — Jul 2022 - Aug 2024
Machine learning engineer on robustness and scientific machine learning projects

Research Projects

Coarse-to-fine Diversity Reward for Multi-turn Jailbreak Attacks — Feb 2026 - Present
- Injecting strategic and content diversity into attacker training with GRPO.
- Showing that higher diversity improves pass@k and weak-to-strong transferability of attacks.

An Automated Generation Framework for Task-Specific Agentic Safety — May 2026 - Present
- Proposing an automatic generation pipeline that generates task-dependent attacks from policy-grounded risk specifications, with fine-grained prompt-injection locations and templates.

Multi-turn Jailbreak Benchmark for Stress-testing LLM Safety — May 2025 - Jan 2026
- Built a scalable multi-turn jailbreak benchmark with active-learning–based data expansion across diverse harm categories.
- Stress-tested LLMs' safety performance, revealing persistent vulnerabilities to conversational attacks.

PDE Control Using Formal Languages and Large Language Models — Oct 2024 - Jan 2025
- Used LLMs to convert PDE-related natural language into formal temporal logic and executable control code.
- Designed a back-translation and auto-verification pipeline for data generation.

Operator Learning via Unsupervised Pretraining and Scalable In-Context Learning — Nov 2023 - Jan 2024
- Proposed a data-efficient training pipeline for optimizing efficiency and accuracy.
- Studied the limitations on data (masking, subsample ratio, etc.) for the downstream tasks on different PDE systems.

Propagating Geometric Uncertainty through Numerical PDE Solutions — Aug 2024 - Apr 2026
- Built a physics-informed metric to evaluate next-best-view (NBV) methods for 3D reconstruction.
- Propagated geometric uncertainty through differentiable grid-based methods.

IARPA TrojAI for Model Robustness against Backdoor Attack | *contest leaderboards* — Oct 2021 - Aug 2024
- Evaluated model robustness on backdoor attacks using weight-based analysis, trigger inversion, and input perturbation.
- Developed detection and mitigation pipelines for backdoor attacks across domains (CV, LLM, cyber security).
- Placed top 2 in the majority of TrojAI rounds and top 5 in NeurIPS TDC 2022 (detection track).

Publications

MultiBreak: A Scalable and Diverse Multi-turn Jailbreak Benchmark for Evaluating LLM Safety — `arxiv:2605.01687`
J. Song, X. Liu, W. Yang, W. Chen, M. Feng, X. Zhu, J. Gao (*ICML, 2026*)

PDE-Controller: LLMs for Autoformalization and Reasoning of PDEs — `arxiv:2502.00963`
J. Song*, M. Soroco*, M. Xia, K. Emond, W. Sun, W. Chen (*ICML 2025*)

Data-Efficient Operator Learning via Unsupervised Pretraining and In-Context Learning — `arxiv:2402.15734`
J. Song*, W. Chen*, P. Ren, S. Subramanian, D. Morozov, M. W. Mahoney (*NeurIPS 2024*)

FluidGaussian: Propagating Simulation-Based Uncertainty Toward Functionally-Intelligent 3D Reconstruction
J. Song*, Y. Liu*, M. Chanlatte, R. Chowdhury, R. Desai, W. Chen, D. Martin, M. W. Mahoney (*CVPR, 2026*)

Learning Spatiotemporal Dynamics from Sparse Data via a High-order Physics-encoded Network
J. Song*, P. Ren*, C. Rao*, Q. Wang, Y. Guo, H. Sun, Y. Liu (*Computer Physics Communications, 2025*)

Learning Spatiotemporal Dynamics From Sparse Measurements
J. Song*, Z. Song*, P. Ren, N. B. Erichson, M. W. Mahoney, X. S. Li (*Machine Learning: Science & Technology, 2024*)

Learning earthquake ground motions via conditional generative modeling (*Nature Communications*)
P. Ren, I. Naiman, M. Lacour, R. Nakata, N. Nakata, **J. Song**, Z. Bi, O. A. Malik, D. Morozov, O. Azencot, N. B. Erichson, M. W. Mahoney